A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

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Abstract

Detection methods for incipient stator-winding faults are developed one machine at a time, so how the generator type changes the problem is not quantified, and the detectability measures of fault-detection theory presuppose a model or a trained detector. A detector-free statistic is proposed, the signal-to-drift ratio (SDR): the relative change of a feature between a healthy reference interval and a faulted interval, divided by a quantile of its changes between adjacent healthy intervals. The SDR is the known-onset mean-shift test as a standardised effect size, calibrated on each machine's own healthy non-stationarity and comparable across features, operating points and machines. From it a minimum detectable extent, a detectable share and a signature location follow per machine type, sensor suite and fault class; a decision table selects the cheapest adequate suite with bootstrap confidence; and a transfer test fixes the referencing rule under which a detector transfers between machine types. It is demonstrated on open benchmark data from permanent-magnet, squirrel-cage induction and wound-field synchronous generators under identical inter-turn and inter-winding short-circuits. The same design-level fault is three different physical events; the signature relocates between terminal voltage, controller states and field current with the machine–drive combination, and the two machines on one bench differ through their healthy nulls; voltage transducers lower the permanent-magnet machine's detectable extent and add nothing for the induction ma-

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chine; and a detector transfers only when referenced to each trial's own preceding healthy interval. Results carry bootstrap intervals; code and data are open.

Keywords: condition monitoring, fault detectability, stator winding fault, wind-turbine generators, sensor suite selection, transfer across machines

1. Introduction

Stator-winding insulation faults are the archetypal incipient failure of an electrical machine: a short-circuit across a few turns with subtle signatures that evolves into a machine-destroying fault if it is not detected [1, 2]. In wind turbines, electrical-subsystem failures contribute substantially to downtime [3, 4], and a rich literature detects winding faults in a given machine: motor-current signature analysis [5], negative-sequence current and impedance methods [6, 7], the review literature [8, 9] and, for converter-fed machines, the analysis of how closed-loop current control moves the asymmetry from the currents into the voltage commands [10, 11]. Almost all of it is developed and validated on one machine. Yet the generator type of a large turbine—permanent-magnet synchronous (PMSG), squirrel-cage induction (SCIG), wound-field synchronous (WFSG) or doubly-fed induction—is itself chosen on cost, mass, efficiency and grid-compliance grounds [12, 13], and its condition monitoring is added afterwards from a catalogue of commercial systems [14]. Whether the same fault is equally visible in different generator types, in which measured quantity it appears, and which sensors are therefore worth specifying for each has not been quantified.

Answering such questions requires a measure of detectability with three properties that the existing measures do not combine. It must be computable without a model of the machine and without a trained detector, because the quantity of interest is what the measurements contain, not what one algorithm extracts from them; the detectability and isolability notions of fault detection and isolation (FDI) theory [15–17] and the sensor-placement methods built on them [18] presuppose a model, and a learned detector [19] presupposes data from the very machine, with transfer across machines an open problem [20]. It must be comparable across features in different physical units, across operating points and across machines whose healthy behaviour differs, which rules out signal-to-noise ratios in physical units and the per-feature suitability metrics of prognostics [21, 22]. And it must connect to the

quantities a monitoring specification is written in: the smallest fault that shall be detected, the false-alarm rate tolerated and the sensors it costs.

This paper proposes such a measure, the signal-to-drift ratio (SDR): the relative change of an observable feature between a healthy reference interval and a faulted interval, divided by a quantile of the relative changes the same feature exhibits between adjacent healthy intervals when nothing is wrong. Statistically it is the known-onset, fixed-sample mean-shift test of change detection [23] expressed as a standardised effect size; an SDR of one marks the point at which a fault of a given extent becomes distinguishable from the ordinary non-stationarity of the healthy machine at a chosen per-trial false-alarm rate. Because the reference interval of every trial, faulted or not, contributes to the null, the SDR is calibrated on each machine's own healthy behaviour and needs no dedicated healthy recordings. From the SDR follow the quantities a specification can be written on: the *minimum detectable extent* (MDE) of a fault class for a given machine type and sensor suite, the *detectable share* (DS) of trials above a required extent in which the fault is detected, and the *signature location*, the measurement group in which the fault appears. A decision table over the required extent and the false-alarm level turns them into the selection of the cheapest adequate sensor suite, and a transfer test establishes under which feature-referencing rule a detector built on one machine type serves another.

The question is made answerable empirically by a recent series of open benchmark datasets [24–26] in which three types of generator—a PMSG, an SCIG and a variable-speed WFSG—were subjected to the *same* inter-turn and inter-winding short-circuits, inserted at the same tapped positions of the stator winding; the PMSG and SCIG share one bench, one converter, one controller and one protocol, so that everything but the machine and the control structure it requires is held fixed. The datasets were published for algorithm benchmarking and refrain from comparing the machines; here the comparison is the object.

This paper makes four contributions. The first is the SDR as a detector-free, self-calibrated measure of fault detectability, with its statistical interpretation, its pooled null, a binomial reliability screen that removes features with excess false alarms on healthy trials, and the three derived quantities MDE, DS and signature location (Section 3). The second is a six-step procedure that turns these into a selection: a decision table that classifies every sensor suite of every machine type as meeting, failing or uncertain against a requirement on the extent and the false-alarm level, with bootstrap confi-

dence, and a transfer test over feature-referencing rules for a detector that must serve several machine types. The third is a quantitative comparison, the first known to the author, of winding-fault detectability across three wind-turbine generator types under identical faults, with four findings: the design-level extent of a fault and its physical severity are different axes and must both be carried; the measurement group in which a fault becomes visible depends on the machine–drive combination, and the detectability difference between the two machines on one bench comes from their healthy nulls rather than from their signatures; the cheapest adequate sensor suite depends on the machine type *and* on the requirement; and a learned detector transfers across the machines only when its features are referenced to each trial's own immediately preceding healthy interval, not to a commissioning baseline. The fourth is a sensitivity and uncertainty analysis of the tuning choices (null quantile, window length, screen), with trial-level and tap-pair-level bootstrap intervals, and open code that reproduces every result from the public data.

The paper is organised as follows: Section 2 positions the SDR among the detectability measures of FDI and health monitoring and summarises what is known about winding-fault signatures in converter-fed machines; Section 3 defines the SDR, the derived quantities and the six-step procedure; Section 4 describes the three machines, the fault protocol, the sensor suites and the computation; Section 5 reports the results; Section 6 draws out the implications for the condition monitoring of wind-turbine generators and the limits of what one machine per type supports; and Section 7 concludes. Appendix A gives feature definitions, computation details and four additional tables.

2. Background and related work

2.1. Detectability measures in fault detection and health monitoring

Fault detection and isolation has a mature theory of diagnosability for systems described by models. Structural analysis determines which faults can be detected and isolated from which sensor sets without numerical parameter values [16, 17], sensor placement selects the cheapest sensor set that makes a required fault set detectable [18], and the approach has been applied to electric-vehicle drives with a diagnosability index that ranks candidate residual generators [27]; testability analysis in the test-engineering tradition records the same information as a dependency matrix of tests against faults

[28, 29]. Basseville [15] gives statistical definitions of detectability and isolability for parametric models as distances between the distributions of a residual under the healthy and the faulted hypothesis. All of these presuppose a model whose structure is known and trusted [30, 31]. Data-driven diagnosis [19] replaces the model by data from the very machine being monitored, and its transfer across machines is an active problem [20]. The prognostics and health management (PHM) community grades candidate features by monotonicity, trendability and prognosability [21], evaluates prognostic algorithms by dedicated metrics [22] and has reviewed the sensor systems that feed them [32]. None of these yields, for a fault of known extent in a built or simulated machine, a number that states how far above the healthy non-stationarity of each candidate measurement the fault stands, comparable across measurements in different units and across machines.

Table 1 places the SDR among these methods. It requires neither a model structure nor a trained detector, only trials with a known fault extent and observable features, and it produces a magnitude that is comparable across features, operating points and machines because it is normalised by each machine's own healthy non-stationarity rather than by a physical unit. Statistically it is the known-onset, fixed-sample mean-shift test of Basseville and Nikiforov [23] expressed as a standardised effect size (Section 3.2); what is new is its calibration on the reference intervals of every trial, the reliability screen, the limit-of-detection quantities derived from it, and its use to compare machine types and sensor suites and to select among them. It deliberately stops at detectability: isolation, severity estimation and prognosis presuppose it.

2.2. Stator winding faults in converter-fed machines

Inter-turn short-circuits in stator windings are among the most studied incipient faults in electrical machines [1, 2]. Motor-current signature analysis [5], negative-sequence current and impedance methods [6, 7], turn-fault models [33] and the review literature [8, 9] have established the physical signatures: an asymmetry of the winding produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous reference frame. Under closed-loop current control the current regulator acts against the negative-sequence current it can see, so the asymmetry migrates from the currents into the voltage commands and the controller states, with the degree of migration set by the current-loop bandwidth relative to twice the electrical frequency [10, 11].

Table 1: Positioning of the signal-to-drift ratio among detectability and feature-assessment methods

Method	Input	Output	Needs model or detector	Compares machines and sensor suites
Structural diagnosability, sensor placement [16–18]	model structure	detectable and isolable fault sets per sensor set	model structure	per model, no magnitude
Statistical detectability [15]	parametric model, residual	distance between hypotheses	model	per model
Testability analysis [28, 29]	test–fault dependency matrix	detectable and isolable fault sets, test sequences	dependency model	no
PHM feature suitability [21]	run-to-failure data	feature scores	no	no
Learned detectors [19]	labelled data of the machine	detector and its ROC	trained detector	only through transfer [20]
This paper	trials with known fault extent (measured or simulated)	MDE, detectable share and signature location per machine type and sensor suite; decision table; transfer rule	neither	yes, scale-free and self-calibrated

MDE: minimum detectable extent; PHM: prognostics and health management; ROC: receiver operating characteristic.

In a wound-field synchronous machine the stator negative sequence additionally induces a component at twice the electrical frequency in the field current [34], and in permanent-magnet machines with an accessible neutral the zero-sequence voltage is a sensitive indicator [35]. These facts are well known to drive specialists but have not been turned into a measure that can be applied across candidate machine types. Machine-learning detectors are typically developed for one machine [36, 37], and open datasets of winding faults exist for one machine type at a time [38]; the transfer experiment in this paper is the cross-machine counterpart of such detectors. Finite-element turn-fault models [39] and digital twins [40] are the natural trial generators for an instantiation of the method before hardware exists.

2.3. Generator type as a design decision

The generator concept of a large wind turbine—direct-drive or geared; permanent-magnet, induction, wound-field or doubly-fed; full or partial

power conversion—is selected early on cost, mass, efficiency and grid-compliance grounds [12, 13]. Reliability enters the trade through failure-rate statistics of subassemblies [3, 4], and condition monitoring is added afterwards from a catalogue of commercial systems built around vibration, oil and electrical signals [14], not through the observability of the failures in each concept. The design community's concept-stage failure reasoning—function–failure mapping and functional failure identification and propagation [41–43]—and the detection rating of a design failure mode and effects analysis record which failures a concept can suffer and an expert's judgement of whether they would be detected, not a measured magnitude. Because monitoring designs increasingly serve a family of variants [44], whether a monitoring design transfers across machine types is a family-level question, which the transfer step of the method addresses.

3. Method

3.1. Definitions and terms

Let $\mathcal{A}$ be a set of candidate *alternatives*. An alternative $a \in \mathcal{A}$ is a machine type together with the control structure it implies—for a generator, the machine topology and the converter control that the topology requires. Let $\mathcal{F}$ be a set of failure-mode classes; each class $f \in \mathcal{F}$ is parameterised by an *extent* $e \in E_f$ expressed at the design level, so that the same (f, e) can be realised in every alternative: for a winding fault the fraction of a winding branch that is short-circuited, for a bearing a defect size, for a structure a crack length. The design-level extent does not fix the physical severity of the event (Section 5.1), so the framing also names a *physical severity axis* on which the alternatives can be compared. Let $\mathcal{S}$ be a set of candidate *sensor suites*; a suite $s \in \mathcal{S}$ is a set of measurements, including quantities internal to the drive controller, with a cost $c(s)$, and $\Phi(s)$ is the set of features computable from those measurements alone. Let $\mathcal{O}$ be the operating envelope, discretised into operating points o.

A *trial* is a record of the system in alternative a at operating point o containing a reference interval R in which the system is healthy and, immediately after it, a faulted interval F in which failure mode (f, e) is present. Trials may come from a physical test (test-stage instantiation) or from a validated simulation (design-stage instantiation, Section 3.5). These terms—alternative, trial, sensor suite, extent (design level) and severity (physical)—are used throughout.

3.2. The signal-to-drift ratio

The reference interval is split into two adjacent halves R_1, R_2, with R_2 adjacent to F. For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_{R_1}, \bar{\varphi}_{R_2}, \bar{\varphi}_F$ be the feature's mean values over the intervals, and define the relative change under fault, referenced to the half adjacent to the fault, and the relative change between the two healthy halves,

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_{R_2}}{|\bar{\varphi}_{R_2}|}, \qquad \delta^0_\varphi = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}. \tag{1}$$

δ^0_φ measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. Because every trial has a healthy reference interval, it is available from every trial of alternative a, faulted or not, and its absolute value pooled over the N_a trials forms an empirical null distribution of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\mathrm{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha\big(|\delta^0_\varphi|\big)}, \tag{2}$$

where q_α is the α-quantile of the pooled null; with $\alpha = 0.95$, $\mathrm{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable interval pairs.

What the SDR is statistically. For a roughly Gaussian null, $q_{0.95}(|\delta^0|) \approx 1.96\,\sigma$, so $\mathrm{SDR} \geq 1$ is a two-sided 5 % test on the shift of interval means with a known change time: the fixed-sample mean-shift test, the known-onset special case of the generalised likelihood ratio of Basseville and Nikiforov [23]. The SDR is thus a trial-level standardised effect size (Cohen's d on interval means, divided by 1.96), the detectable share defined below is an empirical power, and a share at a given α is one point of a receiver operating characteristic. Three assumptions are built in: the onset is known, the whole faulted interval is averaged, and the healthy reference lies immediately before the fault; the last makes the null one of sub-second non-stationarity, and Section 5.7 quantifies how much wider the null of a reference established at commissioning is. The SDR characterises the simplest detector on the feature—a threshold on the relative change of interval means—and the design quantities derived from it are statements about that detector; a detector that combines features or accumulates evidence over time can detect below

SDR = 1, which is why the oracle variant and the transfer experiment of step 6 are part of the method.

Four properties make the SDR suitable as a measure for comparing alternatives and suites: it is *scale-free*, so a current ratio, a voltage harmonic and a controller output are compared on one axis; *calibrated* against each alternative's own non-stationarity, so a noisier alternative is not credited with detectability it does not have, at the price that numerator and denominator must be reported separately when alternatives are compared (Section 5.2), because a quieter null raises the SDR as much as a stronger signature does; *per trial*, so its distribution over trials carries the dependence on operating point and extent; and *detector-free*, so it can be computed before any algorithm is chosen. The relative form presumes a non-zero healthy level; for features whose healthy value is a noise floor an absolute-change variant with the same structure is used, and Section 5.7 shows that the two rank the features alike.

Safeguards. (i) *Reliability screen.* On healthy trials—trials in which the "faulted" interval contains no fault, only the switching operation that would insert one—a feature should exceed SDR = 1 at the nominal rate $1 - \alpha$. A feature whose count of such false alarms in the n_h healthy trials is incompatible with that rate (one-sided binomial test at the 5 % level; three or more of nine at $\alpha = 0.95$) is excluded from the design quantities for that alternative, as are features whose null quantile exceeds one (healthy relative change above 100 %). (ii) *Fixed, nested and oracle feature.* A suite is evaluated by one *fixed* feature—the screened feature with the highest median SDR over the trials of a class, chosen once—or by the *oracle* feature, the best feature of each trial. The fixed variant is what a threshold detector achieves; the oracle variant is an upper bound for a detector that combines the suite's features and, being a maximum over features, is reported with its false-alarm rate on healthy trials. Because the fixed feature is chosen on the trials it is evaluated on, a *nested* variant chooses it on all but one tap pair and evaluates it on the held-out pair. (iii) *Uncertainty.* Every design quantity carries a bootstrap interval: over tap pairs (the physical fault locations) for statements about a class, over operating points within an extent level for the minimum detectable extent, with Wilson intervals for shares.

3.3. Derived quantities

From the SDR distribution over trials the following are defined for each alternative a, suite s and class f:

- the *diagnosability map* $D_a(s, f, e, o)$: the SDR of the suite's fixed feature at extent e and operating point o;

- the *minimum detectable extent* $\mathrm{MDE}_a(s, f)$: the smallest tested extent level e^* such that $\mathrm{median}_o\, D_a(s, f, e, o) \geq 1$ for every tested $e \geq e^*$ (limit-of-detection convention; equal extents at different winding locations form one level);

- the *detectable share* $\mathrm{DS}_a(s, f \mid e \geq e_{\mathrm{req}})$: the fraction of faulted trials with $e \geq e_{\mathrm{req}}$ in which SDR ≥ 1, written for a required extent so that it does not depend on how the test matrix samples the extent axis;

- the *signature location*: the measurement group—stator currents, terminal voltages, controller internals, mechanical quantities, excitation—whose best screened feature attains the highest SDR in the largest share of trials.

The *alternative-dependent diagnosability* of a set of alternatives is then the map that assigns to every triple (a, s, f) the triple $(\mathrm{MDE}_a, \mathrm{DS}_a, \mathrm{location})$. For a requirement $(f, e_{\mathrm{req}}, \alpha)$ the *adequate* suites of alternative a are those with $\mathrm{MDE}_a(s, f) \leq e_{\mathrm{req}}$, and diagnosability is alternative-dependent for that requirement when the cheapest adequate suite differs between alternatives by more than the bootstrap uncertainty of the MDE. MDE and DS are the quantities against which requirements are written ("inter-turn faults of extent $\geq e_{\mathrm{req}}$ shall be detectable over the operating envelope at a per-trial false-alarm rate $\leq 1 - \alpha$"); the signature location tells the designer *why* an alternative behaves as it does.

3.4. Procedure

The procedure (Fig. 1) has six steps. Steps 3–4 are mechanical and fully specified by Eqs. (1)–(2); steps 1, 2, 5 and 6 are design decisions, for which the method supplies default rules.

1. **Frame.** Enumerate $\mathcal{A}$, the classes $\mathcal{F}$ with an extent scale E_f realisable in every alternative and a physical severity axis, the suites $\mathcal{S}$ and the envelope $\mathcal{O}$. Set e_{req} as the smallest extent whose growth to an unacceptable consequence is faster than the inspection interval, and $1 - \alpha$ as the per-trial false-alarm rate the operator tolerates over that interval. Order the suites by cost: an ordinal rank by added hardware

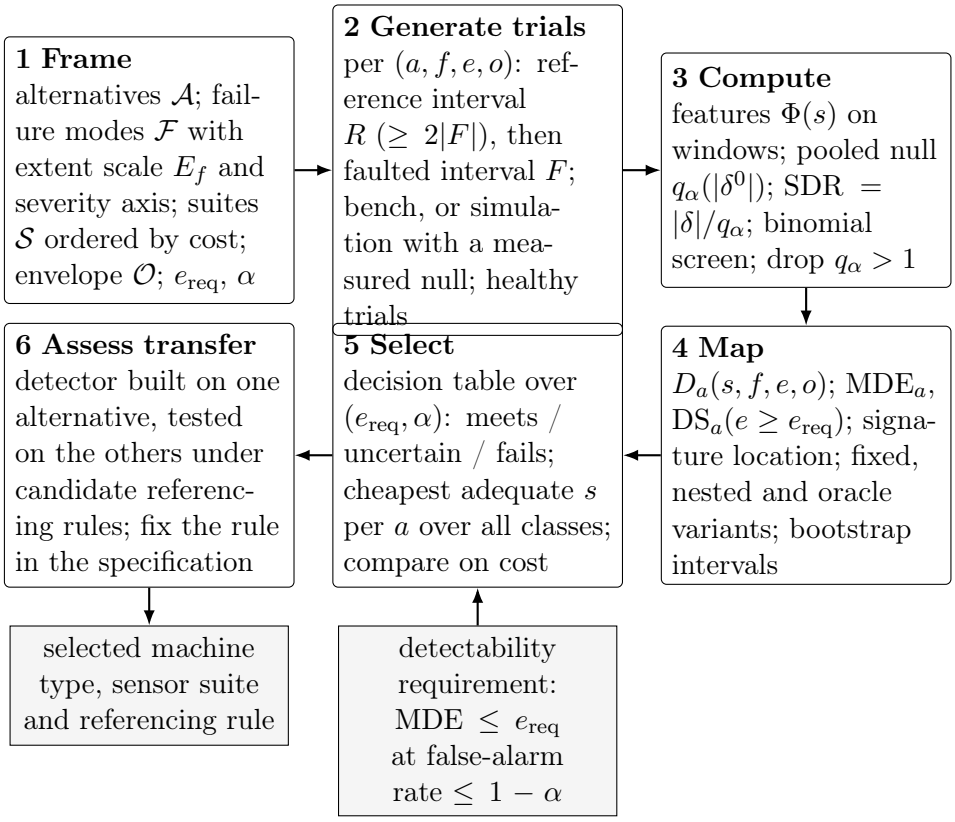


Figure 1: The six-step procedure; steps 3–4 are fixed by Eqs. (1)–(2), steps 1, 2, 5 and 6 follow the default rules of Section 3.4

and integration by default, a monetary cost including installation and maintenance when available.

2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference interval of at least twice the faulted duration, immediately followed by the faulted interval; test at least the extent levels that bracket e_{req} and the corners of the envelope; include enough healthy trials with the same timing for the screen (nine allow the binomial rule above). At the test stage these are prototype or benchmark experiments; at the design stage the faulted numerator is simulated and the null is supplied as in Section 3.5.
3. **Compute.** Extract $\Phi(s)$ for all suites on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, apply the screen.

4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into MDE_a, DS_a and the signature location for the fixed, nested and oracle variants, with bootstrap intervals.
5. **Select.** Build the decision table: for each alternative, class and suite and for the candidate values of e_{req} and α, classify the suite as *meets* when the bootstrap probability that $\mathrm{MDE} \leq e_{\mathrm{req}}$ is at least 0.95, *fails* when it is at most 0.05, and *uncertain* otherwise. The adequate suite of an alternative is the cheapest that meets the requirement for *every* class; an uncertain verdict is resolved by more trials around e_{req} or by specifying the next suite as a fallback. The alternatives are then compared on the cost of their adequate suites together with the other selection criteria; the choice of machine type is affected only when, for some alternative, no affordable suite meets the requirement, otherwise the method changes the monitoring specification and says so.
6. **Assess transfer.** If the suite and its detector must serve a family of alternatives, test whether a detector built on one alternative's trials detects the others' faults under the candidate feature-referencing rules (physical units; scaled on the target's healthy trials; referenced to each trial's own preceding healthy interval). The rule that transfers becomes part of the monitoring specification.

3.5. Design-stage instantiation

The demonstration in this paper is a test-stage instantiation on built machines. A design-stage instantiation, before hardware exists, must supply the two ingredients of the SDR separately. The numerator—the change of a feature under a fault of extent e—is what a finite-element or circuit model of each alternative with the fault injected can provide [39]. The denominator—the healthy non-stationarity of sensor noise, controller activity, thermal drift and operating-point wander—is a property of hardware and environment that such a model does not reproduce, and a simulated SDR that supplied its own null would be optimistic by an unknown factor. The null must therefore come from measured healthy records of hardware of the same class (a predecessor product, a prototype of one alternative, or a benchmark such as the one used here), transferred to the other alternatives on the assumption that the healthy non-stationarity of a feature depends on the sensing and control chain more than on the machine; the assumption is testable by comparing the null quantiles of the alternatives for which hardware exists (Table 4 does so for the two machines on one bench), and a simulated instantiation is validated

by comparing simulated and measured SDR on a few physical trials, the role the three benches would play in a design study of the next generator.

3.6. Scope

The method applies to failure-mode classes that can be seeded at a known extent, at a stationary operating point, with a healthy interval immediately before the fault, in every alternative; the extent scale must be realisable in all of them, and the physical severity axis is declared beside it. It evaluates detectability by a threshold on interval means, not the time to detection, and stops at detection: isolation, severity estimation and remaining-life prediction presuppose detectability and can be built on the same trials. It does not model the cost of missed detection, which enters through e_{req} and α. Run-to-failure data, in which the fault grows gradually and no healthy interval precedes a known onset, do not fit the trial structure and would need a different null.

4. Demonstration on three wind-turbine generators

4.1. Alternatives and data

The three alternatives are laboratory-scale generators of the same power class on instrumented test benches at the University of São Paulo, published as open benchmark datasets (Table 2, Fig. 2). The PMSG [24] and the SCIG [25] are four-pole, 230 V, double-star machines from the same manufacturer, rated 2.5 kVA and 2.5 kW, installed on the *same* bench: the same induction-motor prime mover and frequency converter, the same rapid-control-prototyping platform and field-oriented control (FOC; rotor angle from an encoder for the PMSG, from a flux observer for the SCIG), the same sensors, acquisition (20 kHz, 3 s records), fault-insertion hardware and test script. The variable-speed WFSG [26] is a 2 kVA four-salient-pole machine on an earlier bench of the same group [45] with a three-level converter, FOC with encoder angle, 4 kHz acquisition and 1.3 s records. The PMSG–SCIG pair is therefore a controlled comparison in which everything but the machine and the control structure it requires is held fixed; the WFSG is a third alternative with the same fault protocol on a different bench, analysed with that difference kept explicit. Terminal voltages are measured by each bench's voltage transducers, whose bandwidth the data papers do not document, so the terminal-voltage features carry the sensing chain of their bench as well as the machine.

Table 2: The three alternatives; the PMSG and SCIG share one bench and one protocol, the WFSG bench differs where indicated

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, static exciter
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.3 s
Reference / faulted interval	0.68 s / 0.36 s	0.68 s / 0.36 s	0.49 s / 0.06–0.28 s
Fault impedance	2.6 Ω	2.6 Ω	1, 2.83, 5.66, 11.32 Ω
Operating points	3 speeds × 3 torques	same	716–1910 rpm, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials	216	216	438 (198 at 2.83 Ω)

Ratings as given by the manufacturer (kVA for the synchronous machines, kW for the induction machine). 2L: two-level converter; 3L-NPC: three-level neutral-point-clamped converter; FOC: field-oriented control.

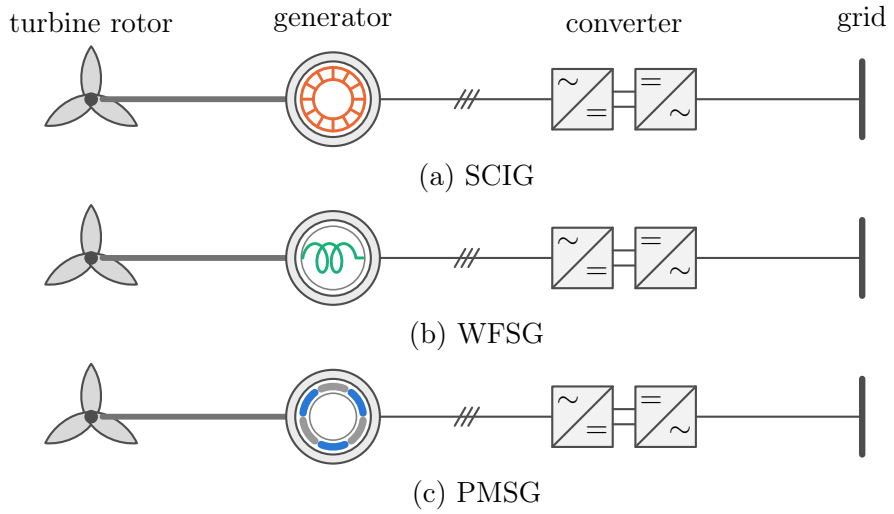
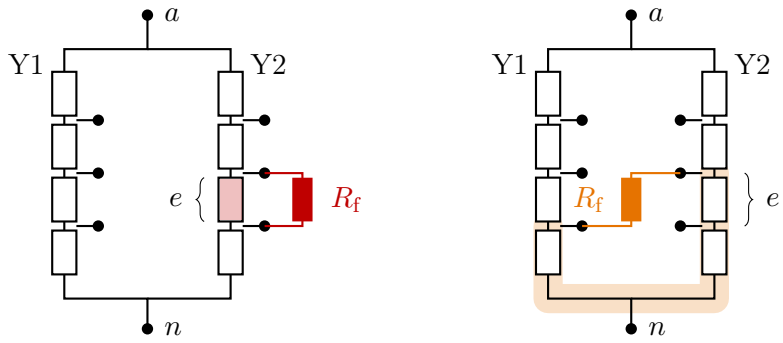


Figure 2: The three alternatives as generator systems: turbine rotor, generator with its rotor excitation (cage, field winding or magnets), converter and grid

4.2. Failure-mode class and extent scale

All three machines were built with 24 tapped derivation points along their stator windings at the same relative positions, so that a short-circuit between two taps realises a fault of known extent; the tap positions are identical in the



(a) Inter-turn short-circuit (b) Inter-winding short-circuit

Figure 3: Inter-turn (**a**) and inter-winding (**b**) short-circuits through R_f between taps of one phase of the double-star stator; e is the fraction of a branch between the taps

three data papers, which is what makes the extent scale realisable in every alternative. Two classes are used (Fig. 3): inter-turn short-circuits (both taps on the same branch) and inter-winding short-circuits (taps on the two parallel branches of a phase). The extent e is the fraction of a branch between the two taps; it ranges from 2.7 % to 12.1 % for the twelve inter-turn cases and from 2.3 % to 40.2 % for the twelve inter-winding cases. Extents that coincide at different winding locations (2.8 % twice, 7.4 % three times, 11.6 % twice) form one extent level. Faults were closed through a contactor and a series resistance after the machine had reached steady state: 2.6 Ω for 0.4 s on the PMSG and SCIG; on the WFSG the experimenters raised the resistance with extent (1 Ω for the inter-turn cases below 11 %, 2.83 Ω for the remaining inter-turn cases and the small inter-winding cases, 5.66 and 11.32 Ω for the large inter-winding cases), so that on that bench extent and impedance are confounded. The fault current is measured on all benches; it is used only as ground truth for the physical severity axis, never as a feature. Because the series resistance is of the order of the winding impedance of these small machines, the fault current is essentially the voltage induced in the shorted turns divided by the resistance: a *resistor-limited* severity, unlike a bolted fault, which the leakage reactance of the shorted turns limits (Section 5.1).

The PMSG and SCIG matrix is complete: 24 fault cases × 3 speeds (1200, 1500, 1800 rpm) × 3 torques (5.2, 6.4, 8.0 N m), the torque–speed pairs scaled from the operating region of the International Energy Agency (IEA) 15 MW reference turbine [46] to the laboratory speed range (the electrical frequency

is not scaled), plus the healthy condition at each operating point; the SCIG generates with slip, at 59.6 Hz at 1800 rpm. The WFSG matrix covers the same 24 tap pairs at six speeds (716–1910 rpm) and four per-unit torque levels, sparsely and with two to six repetitions, so its operating points are not matched to the other two. The WFSG benchmark also contains phase-to-phase faults, a class absent from the PMSG/SCIG protocol; they are excluded from every analysis. The analyses use the WFSG's 438 inter-turn and inter-winding trials at all impedances for the null, the feature ranking and the detectable shares, with the impedance as a covariate, and the 198 trials at 2.83 Ω—the impedance closest to the PMSG/SCIG value, covering four inter-turn tap pairs of 11.6–12.1 % and four inter-winding pairs of 2.3–10.6 %—for the minimum detectable extent and the decision table.

4.3. Sensor suites and features

Five sensor suites are considered, ordered by the data access, hardware and integration they add to a bare generator–converter set (Table 3). *Drive-internal* uses only quantities the converter controller already has; *3 CT* (three current transducers) uses the three phase currents—from dedicated transducers or from the converter's own current sensors routed to a monitoring system—with angle-free features, so that for a converter-fed machine its cost is data access and bandwidth rather than hardware; *3 CT + 3 VT* adds three terminal voltage transducers (VT); *3 CT + 3 VT + drive* adds controller access, which also allows the measured terminal voltages to be transformed into the synchronous frame; and *+ torque transducer* adds a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite), available because its exciter is static [34]. Each feature is assigned to the cheapest suite whose measurements suffice to compute it. The suites are those the datasets can populate (zero-sequence voltage, stray flux, winding temperature, vibration and partial discharge are not recorded, and the angle-free Park's-vector features were not computed), so the suite conclusions hold among the suites and features considered; the zero-sequence current is not used because the star points are isolated.

Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle; the electrical frequency and the phase rotation are estimated per window from the current space vector, so that the external suites need no angle sensor. Appendix A.1 defines every feature and states which WFSG channels are measured and which commanded.

Table 3: Sensor suites and the features they can compute

Suite	Added access / hardware	Features
Drive-internal	controller data	I_d, I_q: std, $1f_e$, $2f_e$; $V_d^{\mathrm{cmd}}, V_q^{\mathrm{cmd}}$: $2f_e$; PI outputs d, q: std, $2f_e$; duty-cycle negative sequence D_2/D_1; T_e: std, $1f_e$, $2f_e$; DC-link std; speed std
3 CT	three phase currents, no angle	I_2/I_1, RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q: $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m: std, $1f_e$, $2f_e$
Excitation (WFSG)	field-current measurement	I_f: std, $2f_e$

f_e: electrical frequency; $1f_e$, $2f_e$: amplitude of the component at once and twice the electrical frequency; I_1, I_2 (V_1, V_2): positive- and negative-sequence magnitudes at f_e; std: standard deviation; THD: total harmonic distortion; PI: proportional–integral.

4.4. Computation

For the PMSG and SCIG the reference interval is the 0.68 s before the contactor command, split into R_1 (0.32 s) and R_2 (0.36 s), and the faulted interval runs from 30 ms after the command to 10 ms before its release; for the WFSG the fault interval is located from the measured fault current and the reference interval (0.49 s) is split at 0.30 s. Appendix A.2 lists all interval boundaries, thresholds, seeds and model settings. The main PMSG–SCIG comparison uses $n_c = 5$; the three-alternative comparison uses $n_c = 3$ for all three machines because the WFSG fault intervals are short. The sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and to $\alpha \in \{0.90, 0.95, 0.99\}$ is reported, with the screen applied at each setting. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 WFSG records, every record contributing a healthy reference interval. Uncertainty is quantified by bootstrap with 2000 resamples: the median SDR and the detectable share of a class are resampled over the twelve tap pairs, the same pairs for both machines so that the SCIG/PMSG ratio is matched; the minimum detectable extent is resampled over the operating-point trials within each extent level; shares also carry Wilson intervals; the null quantile is held at its full-sample value. The decision table uses 1000 resamples per suite, alternative, class and α, with $e_{\mathrm{req}} \in \{3, 5, 8, 12\}$ %.

Step 6 of the procedure is instantiated as a window-level change detector: a gradient-boosted decision-tree (GBDT) classifier [47] and, for the PMSG–SCIG pair, a logistic regression, trained to separate faulted windows from healthy ones. Within an alternative the detector is evaluated by five-fold cross-validation grouped by tap pair (healthy trials grouped by speed), so that the tested fault locations are never seen in training; across alternatives it is trained on all trials of one machine and tested on all trials of another. Three feature-referencing rules are compared: physical units; features z-scored on the target machine's healthy trials (a commissioning baseline); and the absolute z-score of each window against the *first half* of its own trial's reference interval. Negatives are the reference and recovery windows of all trials plus the fault-interval windows of the healthy trials, the only negatives that contain the contactor operation without a fault. Because the windows used to normalise a trial are trivially separable under the third rule, every metric is also reported on the out-of-reference windows, and the false-alarm rate of the threshold that gives 1 % false positives on all negatives is reported on the healthy trials' fault-time windows; the true-positive rate at 1 % false-positive rate is a point of the receiver operating characteristic, not an operating threshold.

All computations are implemented in Python and released with the paper; the pipeline runs from the feature caches to every table and figure in under thirty minutes on a four-core machine. A large language model assisted in writing the analysis code and in drafting and editing the text; every analysis and reference was verified by the author (see the declarations).

5. Results

5.1. The same design-level fault is not the same physical fault

For the same tap pair the SCIG's fault current is about 10 % lower than the PMSG's (medians 3.3 against 3.7 A for inter-turn and 7.7 against 8.5 A for inter-winding pairs), but the difference that matters for detection is relative to what the stator sensors see: the fault current is 1.45 times the pre-fault stator current (RMS to RMS, median over pairs) in the PMSG and 0.63 times in the SCIG for inter-turn faults, and 3.3 against 1.5 times for inter-winding faults, because the SCIG's stator current carries the magnetising component (5.3 against 2.6 A RMS before the fault). The WFSG trials at the matching impedance give 1.20 (inter-turn, 11.6–12.1 %) and 1.03 (inter-winding, 2.3–10.6 %), and 0.65 at the lower impedance used for its small

Table 4: SCIG/PMSG ratio of median SDR decomposed into numerator and denominator, five-cycle windows

Class	Feature	$\lvert\delta\rvert$ P	$\lvert\delta\rvert$ S	q_{95} P	q_{95} S	Num.	Den.	SDR ratio
inter-turn	V_2/V_1	0.257	0.136	0.086	0.033	0.53	2.64	1.40
inter-turn	I_2/I_1	0.238	0.152	0.162	0.034	0.64	4.82	3.08
inter-turn	$I_d\,2f_e$	0.291	0.107	0.563	0.016	0.37	35.09	12.95
inter-turn	$\mathrm{PI}_d\,2f_e$	0.289	0.107	0.557	0.016	0.37	34.71	12.91
inter-turn	$V_q\,2f_e$	0.222	0.155	0.165	0.060	0.70	2.73	1.91
inter-turn	$V_q^{\mathrm{cmd}}\,2f_e$	0.197	0.139	0.162	0.039	0.71	4.17	2.94
inter-turn	D_2/D_1	0.251	0.111	0.090	0.010	0.44	8.96	3.98
inter-winding	V_2/V_1	5.318	1.569	0.086	0.033	0.30	2.64	0.78
inter-winding	I_2/I_1	1.833	1.869	0.162	0.034	1.02	4.82	4.91
inter-winding	$I_d\,2f_e$	2.335	1.188	0.563	0.016	0.51	35.09	17.85
inter-winding	$\mathrm{PI}_d\,2f_e$	2.344	1.188	0.557	0.016	0.51	34.71	17.60
inter-winding	$V_q\,2f_e$	3.688	1.495	0.165	0.060	0.41	2.73	1.11
inter-winding	$V_q^{\mathrm{cmd}}\,2f_e$	0.845	1.184	0.162	0.039	1.40	4.17	5.84
inter-winding	D_2/D_1	2.942	0.854	0.090	0.010	0.29	8.96	2.60

Num.: SCIG/PMSG ratio of the median relative change under fault $\lvert\delta\rvert$; Den.: ratio of null quantiles $q_{95}^{\mathrm{PMSG}}/q_{95}^{\mathrm{SCIG}}$; P: PMSG; S: SCIG.

inter-turn extents. Being resistor-limited (Section 4.2), these currents scale with volts per turn and speed; a bolted fault would reach several times rated current and—the first-order difference between the topologies—could be de-excited in the SCIG and the WFSG but not in the PMSG. A fault specified as "e % of a winding" therefore arrives at the sensors of the three alternatives as three different physical events; Section 5.3 uses both axes.

5.2. Where the signature appears, and why the SDR differs

The null of the SDR differs by up to a factor of 35 between the PMSG and the SCIG (Table 4, columns q_{95}), and not only in sensed quantities: the null of the duty-cycle negative sequence, a controller output that involves no sensor, is nine times larger on the PMSG, so the healthy non-stationarity is a property of the machine–drive combination, not of the transducers alone. The table also decomposes the SCIG/PMSG ratio of median SDR: for inter-turn faults the PMSG's relative changes are the *larger* ones on every feature (SCIG/PMSG ratio of median $\lvert\delta\rvert$ 0.37–0.71), and the SCIG's higher SDR is entirely a quieter null (ratio of null quantiles 2.6–35); for inter-winding faults the numerators are comparable (0.3–1.4) and the denominators again decide.

Fig. 4 shows the median SDR of every feature for the three alternatives (the WFSG pooled over its fault impedances; features to the right of the

dashed line at SDR = 1 are detectable in more than half of the trials). In the PMSG the terminal-voltage group carries the strongest signature in 67 % of inter-turn and 88 % of inter-winding trials, through V_2/V_1 (median SDR 3.0 and 62 at five-cycle windows; 76 and 94 % of trials detectable), and the stator-current negative sequence I_2/I_1 is weak for inter-turn faults (median SDR 1.5, 56 %). In the SCIG the controller group is strongest in 56 and 67 % of trials, through the $2f_e$ ripple of the d-axis current and its PI output (6.7 and 74; 79 and 100 %), the stator-current group in 28–33 % and the terminal-voltage group in 13 % and none. The two machines rank the 30 common features with a Spearman correlation of 0.70 (inter-turn) and 0.89 (inter-winding): the same features are informative in both, in a different order. In the WFSG the strongest inter-turn features are V_2/V_1, the q-axis current ripple and the commanded q-axis voltage, and for inter-winding faults the q-axis ripple, the commanded voltage, the duty cycles and the third current harmonic; its field-current $2f_e$ ripple, an observable the other two machines lack, detected 86 % of the inter-turn trials at the matching impedance and all inter-winding trials (Table A.1). The physical quantity behind the relocation is the incremental negative-sequence impedance $\Delta V_2/\Delta I_2$ seen at the terminals under inter-winding faults: 15 [11, 18] Ω in the PMSG, 7.0 [6.6, 7.5] Ω in the SCIG and 11 [10, 13] Ω in the WFSG (medians and interquartile ranges over trials, three-cycle windows), i.e. the PMSG drive converts a given negative-sequence current disturbance into twice the terminal voltage that the SCIG drive does; for inter-turn faults the PMSG's negative-sequence current increments are of mixed sign with a median indistinguishable from zero, which is why current-side features see little there. Both machines run the same field-oriented controller but differ in angle source (encoder against flux observer), negative-sequence impedance and feature baselines (healthy medians of I_2/I_1, V_2/V_1 and D_2/D_1: 0.0055, 0.0009 and 0.0014 on the PMSG against 0.0079, 0.0020 and 0.0036 on the SCIG, and 0.0148, 0.0073 and 0.0039 on the WFSG), so the relocation belongs to the machine–drive combination, not to the control structure alone.

5.3. Minimum detectable extent

Table 5 gives the minimum detectable extent with its bootstrap interval, the detectable share with its Wilson interval and the median SDR with its tap-pair bootstrap interval for four representative features; Fig. 5 plots the SDR against the design-level extent (left; dots are trials, large markers the median per extent level, their shape the fault impedance) and against the

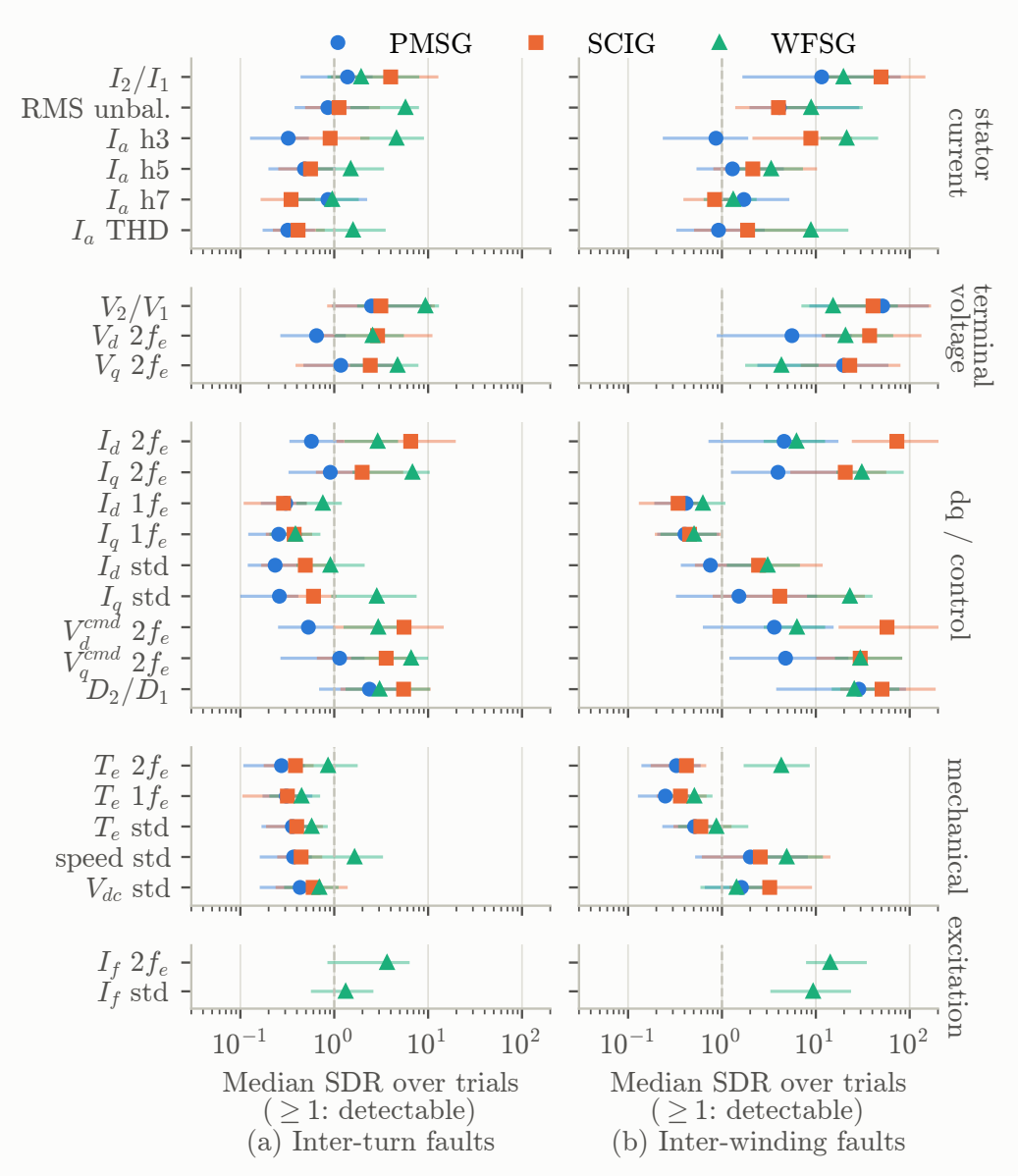


Figure 4: Median SDR of every feature (dot) with interquartile range (bar) for the three alternatives, three-cycle windows

physical severity axis (right; RMS fault current relative to the RMS pre-fault stator current, all trials). Through the terminal-voltage negative sequence

the PMSG detects inter-turn faults from 3.8 % [2.8, 3.8] and the SCIG from 7.4 % [7.4, 7.4]; through the stator-current negative sequence both need 7.4 %, and through the d-axis ripple the PMSG needs 12.1 % and the SCIG 7.4 % [3.8, 7.4]. The smallest tested inter-turn level (2.7–2.8 %) is not reliably detectable on either machine by any single feature. For inter-winding faults the SCIG detects from the smallest tested extent (2.3 %) on every feature except the q-axis voltage ripple, while the PMSG needs 2.3 % [2.3, 14.3] on V_2/V_1 but 9.7 % on I_2/I_1 and 21.7 % on the d-axis ripple. The SDR rises with extent in every machine (Spearman 0.51–0.95), and the MDE intervals are wide because the tested extents form three clusters (about 3, 7.5 and 12 % for inter-turn faults), so the SCIG/PMSG ratio of median SDR is the better basis for comparison: 3.1 [1.3, 5.6] for I_2/I_1 and 12 [2.6, 30] for the d-axis ripple on inter-turn faults, but 1.1 [0.4, 4.6] for V_2/V_1 and 1.5 [0.5, 3.2] for the q-axis voltage ripple. The two machines differ where the signature relocates and are indistinguishable where it does not. On the physical axis (Fig. 5, right) the picture is the same and independent of the tap-pair design: at equal fault-to-stator current ratio the SCIG's current-side features lie above the WFSG's, which lie above the PMSG's, while on V_2/V_1 the three machines nearly coincide. For the WFSG, whose small inter-turn extents were tested only at the lower impedance, the design-level MDE at the matching impedance is bounded by the smallest extent tested there (11.6 %, detectable in 78–98 % of trials depending on the suite); its 2.7–7.7 % inter-turn trials at 1 Ω sit at SDR ≈ 1 on V_2/V_1 and I_2/I_1, and all its inter-winding extents from 2.3 % are detectable.

5.4. Operating envelope

Speed dominates the operating-point dependence: in the PMSG the median SDR of the best feature rises from 5–9 at 1200 rpm to 16–21 at 1800 rpm and the detectable share from 79–83 to 92–96 %, in the SCIG from 16–18 to 44–51 and from 92–96 to 96–100 %; torque changes little. Two cautions apply: the dependence is consistent with a resistor-limited fault current that scales with speed, so it is a property of the emulation as much as of the machines; and the pooled null is a mixture over operating points (the share of trials whose healthy change exceeds the pooled 95th percentile is 0–16 % by speed for the key features), so the 5 % false-alarm reading holds on average over the envelope. Both make the lowest monitored speed a parameter of the requirement.

Table 5: Minimum detectable extent, detectable share and median SDR with 95 % intervals for four features, PMSG and SCIG, five-cycle windows, $\alpha = 0.95$

Class	Feature	M.	MDE [CI]	DS [CI]	Median SDR [CI]	S/P ratio [CI]
IT	V_2/V_1	P	3.8 [2.8, 3.8]	76 [67, 83]	2.99 [1.21, 12.5]	
		S	7.4 [7.4, 7.4]	73 [64, 81]	4.17 [0.949, 15.5]	1.1 [0.378, 4.56]
IT	I_2/I_1	P	7.4 [7.4, 12.1]	56 [47, 65]	1.47 [0.601, 2.36]	
		S	7.4 [7.4, 7.4]	75 [66, 82]	4.52 [1.11, 11.7]	3.07 [1.34, 5.63]
IT	$\mathrm{PI}_d\, 2f_e$	P	12.1 [11.6, 12.1]	31 [23, 40]	0.518 [0.389, 1.11]	
		S	7.4 [3.8, 7.4]	78 [69, 85]	6.69 [1.3, 18.3]	11.8 [2.59, 28.8]
IT	$V_q\, 2f_e$	P	7.4 [7.4, 11.6]	57 [48, 66]	1.35 [0.791, 4.58]	
		S	7.4 [7.4, 7.7]	60 [51, 69]	2.58 [0.516, 5.33]	1.49 [0.464, 3.21]
IW	V_2/V_1	P	2.3 [2.3, 14.3]	94 [88, 97]	61.9 [9.24, 181]	
		S	2.3 [2.3, 2.3]	100 [97, 100]	48.2 [15.3, 188]	0.974 [0.696, 1.68]
IW	I_2/I_1	P	9.7 [9.5, 21.7]	81 [73, 88]	11.3 [1.98, 64.3]	
		S	2.3 [2.3, 2.3]	100 [97, 100]	55.6 [12.5, 117]	5.49 [1.75, 13.1]
IW	$\mathrm{PI}_d\, 2f_e$	P	21.7 [9.7, 21.7]	72 [63, 80]	4.21 [1.18, 13]	
		S	2.3 [2.3, 2.3]	100 [97, 100]	74 [22.9, 245]	18.2 [12.3, 27.2]
IW	$V_q\, 2f_e$	P	9.7 [2.3, 14.3]	90 [83, 94]	22.4 [3.24, 62]	
		S	9.7 [2.3, 9.7]	92 [85, 96]	24.8 [7.04, 78.6]	1.13 [0.831, 1.9]

MDE: % of the winding branch between the taps, limit-of-detection convention, bootstrap over operating points within each extent level; DS: %, Wilson interval; median SDR and SCIG/PMSG ratio: bootstrap over tap pairs, the same pairs for both machines. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG. '–': the largest tested extent is not detectable; smallest tested extents 2.7 % (IT) and 2.3 % (IW).

5.5. Sensor suites and the selection

Table 6 gives the suite results for the PMSG and SCIG at five-cycle windows, Table A.1 (Appendix A) for the three alternatives on the common feature set at three cycles, and Table 7 the decision table of step 5. For the PMSG, current transducers alone resolve inter-turn faults from 7.4 % and inter-winding faults from 9.7 %; three voltage transducers take these to 3.8 and 2.3 %, and neither controller access nor a torque transducer improves the fixed-feature result further. The drive-internal suite reaches 7.4 % for inter-turn faults through the duty-cycle negative sequence, a weaker version of the signature the measured terminal voltage carries. For the SCIG the cheapest suites already reach what any suite reaches—7.4 % for inter-turn faults through I_2/I_1 or, with no added sensor, the d-axis PI output, and 2.3 % for inter-winding faults—and additional hardware raises the detectable share rather than the reach. For the WFSG every suite detects all inter-winding faults from 2.3 % and 78–98 % of the inter-turn trials at the matching impedance, and the field current alone detects 86 and 100 %. The

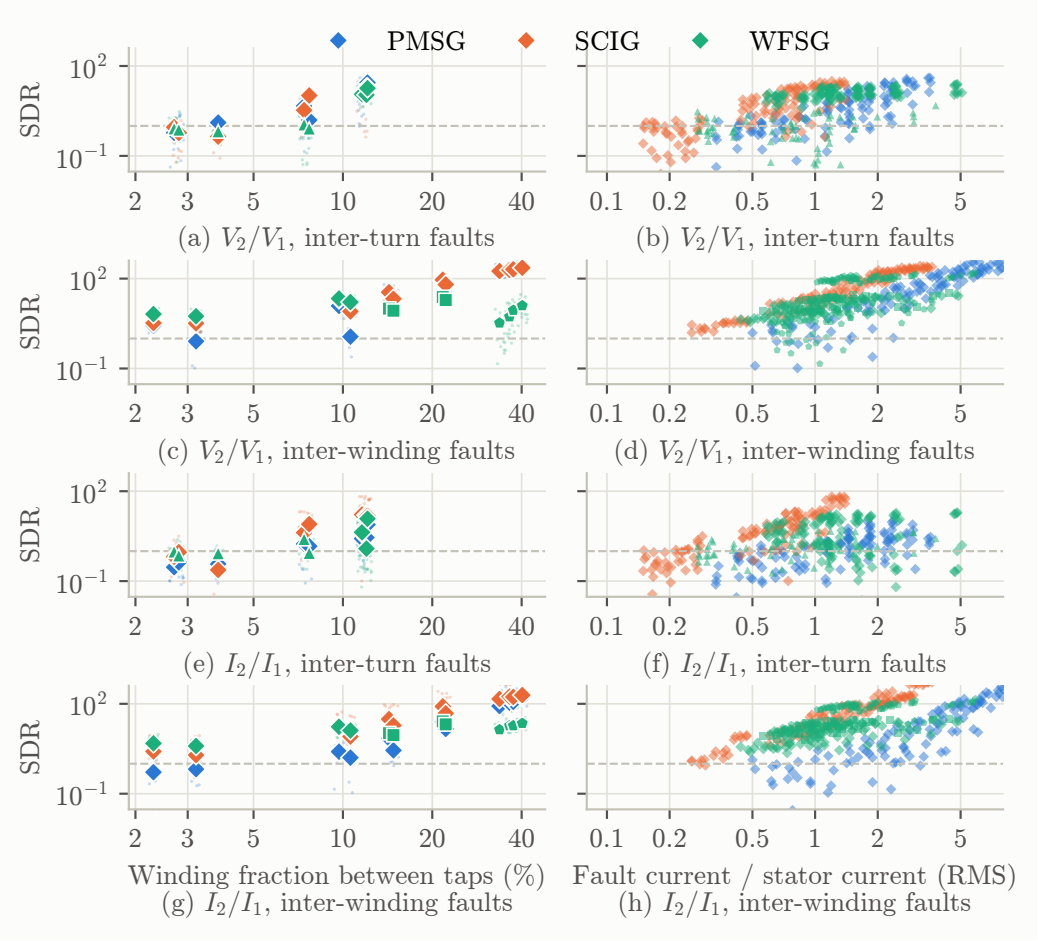


Figure 5: SDR of V_2/V_1 (**a**–**d**) and I_2/I_1 (**e**–**h**) against the design-level extent (left) and the physical severity (right), three alternatives, three-cycle windows

nested selection of the fixed feature changes no share by more than five points (Table A.1); the oracle shares come with false-alarm rates of 44–78 % on the healthy trials for the large suites and are upper bounds only. A shaft torque transducer adds nothing for winding faults on the two machines on which it was available (the torque features' healthy relative change exceeds 100 % on the SCIG and their median SDR is below 0.3 on the PMSG).

The decision table turns this into the selection of step 5. At $\alpha = 0.95$ no suite on any alternative meets an inter-turn requirement of 3 or 5 % with 95 % bootstrap confidence; the PMSG's voltage suite is the only candidate

Table 6: Sensor suites versus what they detect, PMSG and SCIG, five-cycle windows: best fixed feature, its MDE (%) and detectable share (%)

Suite	PMSG feature	MDE	DS [oracle]	SCIG feature	MDE	DS [oracle]
Inter-turn faults						
drive-internal	D_2/D_1	7.4	64 [87]	$\mathrm{PI}_d\, 2f_e$	7.4	78 [99]
3 CT	I_2/I_1	7.4	56 [77]	I_2/I_1	7.4	75 [87]
3 CT + 3 VT	V_2/V_1	3.8	76 [88]	I_2/I_1	7.4	75 [90]
3 CT + 3 VT + drive	V_2/V_1	3.8	76 [95]	$\mathrm{PI}_d\, 2f_e$	7.4	78 [99]
+ torque transducer	V_2/V_1	3.8	76 [95]	$\mathrm{PI}_d\, 2f_e$	7.4	78 [99]
Inter-winding faults						
drive-internal	D_2/D_1	14.3	90 [100]	$\mathrm{PI}_d\, 2f_e$	2.3	100 [100]
3 CT	I_2/I_1	9.7	81 [94]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT	V_2/V_1	2.3	94 [97]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT + drive	V_2/V_1	2.3	94 [100]	$\mathrm{PI}_d\, 2f_e$	2.3	100 [100]
+ torque transducer	V_2/V_1	2.3	94 [100]	$\mathrm{PI}_d\, 2f_e$	2.3	100 [100]

Oracle: per-trial best feature, its detectable share in brackets (false-alarm rates in Table A.1); features failing the reliability screen are excluded for that machine.

below 5 % and its verdict is uncertain, a consequence of the three-cycle windows and the coarse extent grid. At 8 % the drive-internal suite meets the requirement on the PMSG and the SCIG, i.e. no added sensor is needed for either; the WFSG's inter-turn verdicts are bounded by its smallest tested extent. For inter-winding faults the SCIG and the WFSG meet every requirement from 3 % with the drive-internal suite, whereas the PMSG needs current transducers at 12 % and is uncertain below (its five-cycle result, Table 6, reaches 2.3 % with voltage transducers). Relaxing α to 0.90 admits the PMSG's voltage suite at 5 % for inter-turn faults; tightening it to 0.99 removes the PMSG's current-only option at 12 % for inter-winding faults.

5.6. Transfer across alternatives

Table 8 reports step 6 for the PMSG–SCIG pair and Table A.4 (Appendix A) for the three alternatives; the figures quoted below are AUCs on the out-of-reference windows, and the values on all windows agree with them to within 0.02, so the windows used for normalisation do not drive the result. In physical units a gradient-boosted detector reaches an area under the receiver operating characteristic curve (AUC) of 0.87–0.91 within a machine on unseen tap pairs but 0.78 (PMSG→SCIG) and 0.55 (SCIG→PMSG) across; with its features scaled on the target machine's healthy trials, a commissioning baseline, 0.82 and 0.75, and a logistic detector under that rule falls below chance in one direction (0.45); with each feature referenced to the first half

Table 7: Decision table: the cheapest suite whose MDE meets e_{req} with probability ≥ 0.95

α	e_{req} (%)	PMSG		SCIG		WFSG	
		IT	IW	IT	IW	IT	IW
0.90	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.90	5	3CT+3VT	(drive)?	none	drive	none	drive
0.90	8	drive	(drive)?	drive	drive	none	drive
0.90	12	drive	3CT	drive	drive	drive	drive
0.95	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.95	5	(drive)?	(drive)?	none	drive	none	drive
0.95	8	drive	(drive)?	drive	drive	none	drive
0.95	12	drive	3CT	drive	drive	drive	drive
0.99	3	none	none	none	drive	none	drive
0.99	5	(3CT+3VT)?	none	none	drive	none	drive
0.99	8	drive	none	drive	drive	none	drive
0.99	12	drive	(drive)?	drive	drive	drive	drive

Three-cycle windows; probabilities from 1000 bootstrap resamples. Suites ordered drive < 3CT < 3CT+3VT < 3CT+3VT+drv; exc: WFSG field current; IT: inter-turn; IW: inter-winding. '(suite)?': no suite meets with that confidence and the cheapest suite with an uncertain verdict is shown; 'none': every suite fails. WFSG inter-turn verdicts are bounded by its smallest tested extent at the matching impedance (11.6 %).

of the trial's own reference interval, 0.91 and 0.84, against 0.91–0.94 within a machine, and a logistic detector transfers at 0.89 and 0.87. The threshold that gives 1 % false positives on all negatives fires on 0–3 % of the healthy trials' fault-time windows, the only negatives that contain the contactor operation. Between the three alternatives, self-referenced detectors transfer in all six directions with AUC 0.80–0.98 on the out-of-reference windows, and a detector trained on two machines detects the faults of the third with 0.83 (PMSG), 0.88 (SCIG) and 0.97 (WFSG; optimistic, as that bench has no healthy trials). Restricted to each suite's features, every suite transfers at 0.82–0.95 when self-referenced and at 0.62–0.90 under the commissioning baseline.

5.7. Robustness and the two nulls

Table A.2 (Appendix A) shows how the tuning choices move the design quantities, with the screen applied at every setting. The inter-turn MDEs of all five key features are the same at three, five and eight cycles on both machines and the detectable shares move by at most six points; for inter-

Table 8: Transfer of a window-level detector between PMSG and SCIG: AUC on all windows / AUC on the out-of-reference windows

Referencing	Model	P→P	S→S	P→S	S→P
physical units	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.59/0.59
physical units	GBDT	0.88/0.87	0.90/0.91	0.78/0.78	0.55/0.55
z on target's healthy trials	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.44/0.45
z on target's healthy trials	GBDT	0.88/0.87	0.90/0.91	0.82/0.82	0.75/0.75
$\|z\|$ vs own reference	logistic	0.90/0.89	0.94/0.93	0.90/0.89	0.88/0.87
$\|z\|$ vs own reference	GBDT	0.92/0.91	0.95/0.94	0.92/0.91	0.86/0.84

Out-of-reference windows: those outside the normalising set (Section 4.4). P→P, S→S: cross-validation grouped by tap pair; P→S, S→P: trained on one machine, tested on the other.

winding faults one MDE moves (the PMSG's V_2/V_1, 9.7 % at three cycles against 2.3 % at five and eight). Tightening the null quantile from 0.95 to 0.99 lowers the shares by 3–9 points and moves two PMSG inter-winding MDEs up one level; relaxing it to 0.90 raises the shares by up to 9 points and moves one down. No ordering of the two machines reverses, but the PMSG's inter-winding suite conclusion depends on α and on the window length (Table 7). At $\alpha = 0.95$ the screen removes up to three weak features per machine and window length (Appendix A.2); none of the five key features is removed at any setting, and the three-cycle result of V_2/V_1 on the PMSG (two false alarms in nine healthy trials) passes the binomial rule. An absolute-change variant of the SDR, which removes the healthy level of the feature from numerator and null, ranks the 30 features with a Spearman correlation of 0.92–0.98 against the relative variant and picks the same three top features in every machine and class. The bootstrap intervals (Table 5) put the median SDRs within a factor of two to four and the shares within ±8–10 points.

Table A.3 (Appendix A) replaces the reference immediately before the fault by the healthy trial recorded at the same operating point at another time, a commissioning baseline: the null of the four key features widens by a factor of 1.6 to 4.9, the median SDR of the inter-turn features falls to about one (PMSG V_2/V_1 from 3.0 to 1.1; SCIG I_2/I_1 from 4.5 to 1.1) and their detectable shares to 20–59 %, while inter-winding faults remain detectable in 67–94 % of trials. The MDEs reported here are therefore statements about a short-horizon change detector; a monitor that compares against an old baseline needs the wider null.

6. Discussion

6.1. What the demonstration establishes about the method

The demonstration supports four claims about the method that do not depend on which machine came out ahead, and it corrects the naive form of two of them.

The design-level description of a failure mode does not fix its physical severity. A short-circuit of the same winding fraction produced a fault current that differed by about 10 % between the PMSG and the SCIG in amperes and by a factor of 2.3 relative to the stator current the sensors see; on the WFSG bench extent and fault impedance were confounded. A method that compares alternatives on a failure-mode class must therefore carry two severity axes—the design-level one, on which requirements are written, and a physical one, on which the alternatives are compared—and state which one a number refers to; for resistor-limited emulation the physical axis is predictable at the design stage from volts per turn, speed and the resistance.

Where a failure becomes observable is a property of the machine–drive combination, and the SDR must be read with its decomposition. In the PMSG the winding asymmetry appeared in the terminal voltage, in the SCIG in the magnetising-axis current and its controller action, in the WFSG additionally in the field current and the current harmonics; the incremental negative-sequence impedance under fault was twice as large in the PMSG drive as in the SCIG drive. The version of this claim in which "the control structure decides" is not supported: both machines run the same controller, and Table 4 shows that the SCIG's higher SDR on current and d-axis features is entirely a quieter healthy null. The SDR is designed to credit a quiet alternative, but a design team must know whether an alternative wins by signature or by silence, because the two generalise differently: a signature is a property of the fault physics, a null of the sensing and control chain and of the operating regime, which a design-stage instantiation cannot simulate (Section 3.5).

The cheapest adequate sensor suite depends on the alternative and on the requirement, and here it would change the monitoring specification, not the choice of machine type. With three current transducers the PMSG reaches 7.4 % (inter-turn) and 9.7 % (inter-winding); three voltage transducers take it to 3.8 and 2.3 % and buy nothing on the SCIG, whose current-only or drive-internal suites already reach 7.4 and 2.3 %. The decision table makes the requirement explicit: around 8 % of a branch all three alternatives meet it with drive-internal quantities and diagnosability does not discriminate be-

tween them; below 5 % only the PMSG with voltage transducers is a candidate, with an uncertain verdict at the confidence demanded. The method would change the choice of machine type only if, for one alternative, no affordable suite met the requirement; here it changes what is specified for each: a measured detectability per alternative and suite, with its uncertainty, in place of an assumed one.

A monitoring design transfers across alternatives only under the right referencing rule, and the rule is a short-horizon one. A detector built on one machine and applied to another in physical units lost much of its performance; features scaled on the target machine's healthy trials—a commissioning baseline—recovered part of it and were harmful in one direction; features referenced to the first half of each trial's own reference interval transferred in all six directions between the three machines and from any two to the third. The between-trial null (Table A.3) explains why: against a baseline recorded at another time, the inter-turn signatures sit at SDR ≈ 1. For a product family the specification rule is a change detector on self-referenced deviations over a short horizon; a monitor that compares against an old baseline must be calibrated on the wider null.

6.2. Implications for the condition monitoring of wind-turbine generators

Within the scope of Section 6.4, the results carry the following implications for the wind-turbine generator. A PMSG with full-power conversion should be specified with terminal voltage measurement, or with another observable of the terminal asymmetry such as the zero-sequence voltage where a neutral is accessible [35], if inter-turn faults below about 7 % of a branch are to be detected by threshold monitoring; its controller's commanded quantities carry a weaker version of the signature. An SCIG reaches the same detectability with current transducers alone or with no added sensor, and additional hardware buys robustness rather than reach. A wound-field machine with a static exciter offers a further drive-internal channel, the field current, whose $2f_e$ ripple detected every inter-winding fault and most inter-turn faults; a brushless exciter would not expose it [34]. A shaft torque transducer, as filtered on these benches, is not worth specifying for winding faults. Two industrial parameters change these statements and are computable at the concept stage: the ratio of current-loop bandwidth to twice the electrical frequency, which for a direct-drive multi-megawatt PMSG at a fundamental near 10 Hz will push the signature further into the controller and the voltage than in these 40–60 Hz machines; and the grid-connected,

unregulated stator of a doubly-fed generator, where the migration is absent and current-signature routes regain their reach.

6.3. Applying the method to other machines

Nothing in Section 3 is specific to electrical machines, but the scope conditions of Section 3.6 are. A gearbox family would use tooth-damage size as extent, vibration and oil-debris suites, and seeded-damage trials with a healthy run before each damage step; a battery-pack family would use cell-capacity loss, voltage, temperature and impedance suites, and trials in which a degraded cell is switched into a healthy pack.

6.4. Limitations

The most important limitation is that each alternative is represented by one physical machine. The differences reported are differences between three machines; attributing them to topology rather than to the individual units requires replicated machines or simulated populations of each alternative under parameter variation, validated against the measured units. The demonstration is therefore regarded as establishing the method and the mechanisms, not the population-level ranking of the topologies; the two benches also differ in converter, controller, sampling rate and sensing chain.

The fault emulation is resistor-limited and the benches are laboratory scale (2–2.5 kVA); the operating points were scaled from a 15 MW turbine but the electrical frequency and the fault physics were not. Only winding faults and only electrical and controller observables were studied, and the suites are those the datasets can populate (Section 4.3). On the WFSG the fault impedance is confounded with extent; only four tap pairs per class exist at the impedance matching the other two machines, so its inter-turn MDE is bounded, not measured; it has no healthy trials, so the reliability screen cannot be applied to it and its transfer entries lack nuisance-transient negatives; and 81 of its 438 trials hold fewer than three three-cycle windows. The PMSG/SCIG faulted interval contains about 55 ms of healthy time before the relay picks up, which dilutes those SDRs slightly relative to the WFSG's.

Statistically, the pooled null is a mixture over operating points and holds its nominal false-alarm rate on average, not per speed; the null quantile is held fixed inside the bootstrap; the fixed feature is chosen on the trials it is evaluated on, which the nested variant shows to be immaterial here; the oracle variant is a maximum over features and its false-alarm rate is reported for that reason; the minimum detectable extent is a discrete quantity on a

coarse extent grid, so the continuous SDR and its ratio are the better basis for comparing alternatives. Other nuisance transients (load steps, grid events) were not in the data. Finally, the SDR characterises a short-horizon change detector; the between-trial null shows that a monitor referenced to an old baseline faces a null 1.6 to 4.9 times wider on the key features, and every MDE in this paper should be read with that qualification.

7. Conclusions

A detector-free measure of fault detectability has been proposed, the signal-to-drift ratio: a scale-free statistic calibrated on each machine's own healthy non-stationarity, from which a minimum detectable extent and a detectable share above a required extent can be computed for every combination of machine type, sensor suite and failure-mode class, with a reliability screen, bootstrap intervals and a decision table over the required extent and the false-alarm level. A six-step procedure with default rules for its framing decisions turns these into the selection of the cheapest adequate sensor suite for each machine type, a comparison of the machine types on that basis, and a transferability specification for a detector that must serve several of them.

Applied to three types of wind-turbine generator subjected to identical winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault across the machines; that where a fault becomes visible depends on the machine–drive combination, the detectability difference between the two machines on one bench arising from their healthy non-stationarity rather than from the signature; that the cheapest sensor suite meeting a requirement depends on the alternative and on the requirement, voltage transducers lowering the permanent-magnet machine's detectable extent and adding nothing for the induction machine, while around 8 % of a branch all three alternatives are equally diagnosable with drive-internal quantities; and that a learned detector transfers across the three machines only when its features are referenced to each trial's own preceding healthy behaviour. The findings are robust in their orderings to the method's tuning, carry tap-pair-level uncertainty, and are limited by one physical unit per alternative, laboratory scale, resistor-limited fault emulation and a single failure-mode class.

Three extensions follow: the design-stage instantiation, in which simulated faulted trials are combined with measured healthy nulls of hardware of the same class and validated on a few physical trials, so that the method ap-

plies before hardware exists and machine-level differences become population-level statements; other failure-mode classes and observables—bearing and rotor faults with vibration and flux observables, zero-sequence and Park's-vector features for the winding faults studied here—and a mechanical or electrochemical family, to test the claim to generality; and a cost model including integration, maintenance and missed detection in place of the ordinal ranking of suites, so that the co-selection can be posed as an optimisation. The released code, which reproduces every result from the public data, is the starting point for all three.

CRediT authorship contribution statement

Hüseyin Tayyer Canseven: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data are public under the CC BY 4.0 licence: the PMSG, SCIG and synchronous-generator datasets of the InnovaPower MitDev-Eletrica project, https://github.com/InnovaPower/MitDev-Eletrica, archived on Zenodo at https://doi.org/10.5281/zenodo.15741561, https://doi.org/10.5281/zenodo.17161986 and https://doi.org/10.5281/zenodo.13685630. The complete analysis pipeline, with the derived per-trial tables, is released under an open licence at https://github.com/htcanseven/Claude (directory `scripts/`) and reproduces every number in this paper from the public data; an archived release with a DOI will be deposited on Zenodo at acceptance.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used a large language model (Claude, Anthropic) under their direction to assist with code development, literature retrieval, data analysis, and drafting and editing of the text. After using this tool, the author reviewed and edited the content as needed, verified all analyses, checked every reference against its published record, and takes full responsibility for the content of the publication. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A. Feature definitions, computation details and additional tables

This appendix defines the features (Appendix A.1), lists the computation details (Appendix A.2) and gives the sensor-suite results for the three alternatives (Table A.1), the sensitivity of the design quantities to the window length and the null quantile (Table A.2), the commissioning-baseline null (Table A.3) and the transfer results between the three alternatives (Table A.4).

Appendix A.1. Feature definitions

Let x_a, x_b, x_c be the sampled phase quantities of a window of N samples at sampling frequency f_s, w the Hann window and f_e the electrical frequency of the window, estimated from the slope of the unwrapped phase of the current space vector $\frac{2}{3}(i_a + \mathbf{a}\, i_b + \mathbf{a}^2 i_c)$, $\mathbf{a} = e^{j2\pi/3}$, whose sign gives the phase rotation (outside 0.5–1.5 times the nominal value the nominal value is used). The complex fundamental amplitude of a signal x at frequency f is $X(f) = \frac{2}{\sum_n w_n} \sum_{n=0}^{N-1} x_n\, w_n\, e^{-j2\pi f n/f_s}$. Sequence magnitudes are $|X_+| = |X_a + \mathbf{a}X_b + \mathbf{a}^2 X_c|/3$ and $|X_-| = |X_a + \mathbf{a}^2 X_b + \mathbf{a}X_c|/3$ for an abc rotation (interchanged for acb), evaluated at $f = f_e$ for currents (I_1, I_2), terminal voltages (V_1, V_2) and mean-removed duty cycles (D_1, D_2); $|X(f)|$ is a peak amplitude and is divided by $\sqrt{2}$ wherever it is compared with an

Table A.1: Sensor suites for the three alternatives, three-cycle windows, common feature set, $\alpha = 0.95$

Suite	Machine	Feature	MDE	DS	DS nested	Oracle DS	Oracle FA
Inter-turn faults							
drive-internal	PMSG	D_2/D_1	7.4	65	65	82	44
drive-internal	SCIG	$I_d\,2f_e$	7.4	78	78	93	78
drive-internal	WFSG	$I_q\,2f_e$	11.6[a]	91	87	96	–
3 CT	PMSG	I_2/I_1	7.4	57	57	78	11
3 CT	SCIG	I_2/I_1	7.4	70	70	86	56
3 CT	WFSG	I_a h3	11.6[a]	98	86	94	–
3 CT + 3 VT	PMSG	V_2/V_1	3.8	73	73	85	33
3 CT + 3 VT	SCIG	I_2/I_1	7.4	70	70	89	56
3 CT + 3 VT	WFSG	V_2/V_1	11.6[a]	98	98	94	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	3.8	73	68	94	67
3 CT + 3 VT + drive	SCIG	$I_d\,2f_e$	7.4	78	78	97	89
3 CT + 3 VT + drive	WFSG	V_2/V_1	11.6[a]	98	98	98	–
excitation (field current)	WFSG	$I_f\,2f_e$	11.6[a]	86	86	74	–
Inter-winding faults							
drive-internal	PMSG	D_2/D_1	14.3	91	91	98	44
drive-internal	SCIG	$I_d\,2f_e$	2.3	100	100	100	78
drive-internal	WFSG	$I_q\,2f_e$	2.3	100	100	100	–
3 CT	PMSG	I_2/I_1	9.7	83	83	99	11
3 CT	SCIG	I_2/I_1	2.3	98	98	99	56
3 CT	WFSG	I_2/I_1	2.3	100	100	100	–
3 CT + 3 VT	PMSG	V_2/V_1	9.7	92	92	99	33
3 CT + 3 VT	SCIG	I_2/I_1	2.3	98	98	100	56
3 CT + 3 VT	WFSG	V_2/V_1	2.3	100	100	100	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	9.7	92	92	100	67
3 CT + 3 VT + drive	SCIG	$I_d\,2f_e$	2.3	100	100	100	89
3 CT + 3 VT + drive	WFSG	$I_q\,2f_e$	2.3	100	100	100	–
excitation (field current)	WFSG	$I_f\,2f_e$	2.3	100	100	100	–

Fixed feature: highest median SDR among the suite's screened features. MDE (%) at the fault impedance matching the PMSG/SCIG protocol (WFSG: 198 trials, four tap pairs per class). DS: detectable share (%) of the fixed feature; nested: fixed feature chosen on the other tap pairs; oracle: per-trial best feature, with its false-alarm rate on the healthy trials (FA, %; not available for the WFSG). '–': largest tested extent not detectable.
[a] The smallest inter-turn extent tested on the WFSG at that impedance is 11.6 %, so the value is an upper bound.

RMS value. The RMS unbalance is $\max_k |I_{\mathrm{rms},k} - \bar{I}_{\mathrm{rms}}|/\bar{I}_{\mathrm{rms}}$ over the three phases; the harmonic ratios are $|I_a(kf_e)|/|I_a(f_e)|$ for $k = 3, 5, 7$ and the THD is $\sqrt{\sum_{k=2}^{20} |I_a(kf_e)|^2}/|I_a(f_e)|$. For the synchronous-frame currents i_d, i_q and the torques the features are the standard deviation of the mean-removed signal and its amplitudes $|Y(f_e)|$ and $|Y(2f_e)|$; for the PI outputs the standard

Table A.2: Sensitivity of MDE/DS to the window length (3, 5, 8 cycles) and to the null quantile (α 0.90, 0.99)

Class	Feature	Machine	3 c	5 c	8 c	α 0.90	α 0.99
IT	V_2/V_1	PMSG	3.8/73	3.8/76	3.8/73	3.8/80	3.8/69
IT	V_2/V_1	SCIG	7.4/69	7.4/73	7.4/75	7.4/76	7.4/65
IT	I_2/I_1	PMSG	7.4/57	7.4/56	7.4/58	7.4/65	7.4/52
IT	I_2/I_1	SCIG	7.4/70	7.4/75	7.4/75	7.4/79	7.4/68
IT	$I_d\,2f_e$	PMSG	12.1/29	12.1/30	12.1/28	12.1/33	12.1/24
IT	$I_d\,2f_e$	SCIG	7.4/78	7.4/79	7.4/83	7.4/84	7.4/72
IT	$\mathrm{PI}_d\,2f_e$	PMSG	12.1/30	12.1/31	12.1/28	12.1/33	12.1/24
IT	$\mathrm{PI}_d\,2f_e$	SCIG	7.4/78	7.4/78	7.4/83	7.4/84	7.4/72
IT	$V_q\,2f_e$	PMSG	7.4/55	7.4/57	7.4/60	7.4/65	7.7/50
IT	$V_q\,2f_e$	SCIG	7.4/61	7.4/60	7.4/62	7.4/64	7.4/59
IW	V_2/V_1	PMSG	9.7/92	2.3/94	2.3/94	2.3/94	9.7/91
IW	V_2/V_1	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	I_2/I_1	PMSG	9.7/83	9.7/81	9.7/82	3.2/87	9.7/79
IW	I_2/I_1	SCIG	2.3/98	2.3/100	2.3/100	2.3/100	2.3/97
IW	$I_d\,2f_e$	PMSG	21.7/71	21.7/72	21.7/69	21.7/74	21.7/69
IW	$I_d\,2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$\mathrm{PI}_d\,2f_e$	PMSG	21.7/71	21.7/72	21.7/70	21.7/74	21.7/69
IW	$\mathrm{PI}_d\,2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$V_q\,2f_e$	PMSG	9.7/89	9.7/90	9.7/91	9.7/91	9.7/86
IW	$V_q\,2f_e$	SCIG	9.7/88	9.7/92	9.7/88	9.7/95	9.7/84

Cells: MDE (%; limit-of-detection convention; '–': largest extent not detectable) / DS (%). Window-length columns at $\alpha = 0.95$; quantile columns at five cycles. IT: inter-turn; IW: inter-winding.
[a] The feature fails the reliability screen at that setting.

deviation and $|Y(2f_e)|$; for the measured and commanded synchronous-frame voltages $|Y(2f_e)|$; for the field current the standard deviation and $|Y(2f_e)|$; speed and DC-link features are standard deviations. Windows are n_c electrical cycles long with a hop of one cycle, and a feature is undefined for a trial whose faulted interval holds no complete window. On the WFSG bench the measured converter-side phase voltages serve as terminal voltages, the synchronous-frame voltages are obtained from them with the encoder angle, the commanded voltages are the controller's, and the electrical torque is computed from power and speed.

Appendix A.2. Computation details

Intervals. PMSG and SCIG: reference interval 0.30–0.98 s after the start of each record, split at 0.62 s into R_1 (0.32 s) and R_2 (0.36 s); contactor commanded at 1.000 s; faulted interval from 30 ms after the command to 10 ms before its release; the relay picks up 85–90 ms after the command, so

Table A.3: Within-trial versus commissioning-baseline null, five-cycle windows, $\alpha = 0.95$

Class	Feature	M.	q_{95}		Median SDR		DS	
			within	baseline	within	baseline	within	baseline
IT	V_2/V_1	P	0.086	0.315	2.99	1.05	76	50
IT	I_2/I_1	P	0.162	0.268	1.47	0.978	56	48
IT	$\mathrm{PI}_d\,2f_e$	P	0.557	0.870	0.518	0.441	31	20
IT	$V_q\,2f_e$	P	0.165	0.277	1.35	0.872	57	44
IW	V_2/V_1	P	0.086	0.315	61.9	12.5	94	83
IW	I_2/I_1	P	0.162	0.268	11.3	6.34	81	75
IW	$\mathrm{PI}_d\,2f_e$	P	0.557	0.870	4.21	2.4	72	67
IW	$V_q\,2f_e$	P	0.165	0.277	22.4	10.7	90	81
IT	V_2/V_1	S	0.033	0.111	4.17	1.45	73	59
IT	I_2/I_1	S	0.034	0.130	4.52	1.15	75	54
IT	$\mathrm{PI}_d\,2f_e$	S	0.016	0.078	6.69	1.29	78	53
IT	$V_q\,2f_e$	S	0.060	0.131	2.58	1.54	60	58
IW	V_2/V_1	S	0.033	0.111	48.2	16.1	100	94
IW	I_2/I_1	S	0.034	0.130	55.6	12.9	100	86
IW	$\mathrm{PI}_d\,2f_e$	S	0.016	0.078	74	16	100	88
IW	$V_q\,2f_e$	S	0.060	0.131	24.8	11.2	92	86

Four key features. Within: reference immediately before the fault; baseline: the healthy trial at the same operating point, recorded at another time. DS: detectable share (%). IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

the faulted interval holds about 55 ms of healthy time (at most 15 % of its length), which dilutes the fault-interval mean and makes the PMSG/SCIG SDRs slightly conservative; recovery windows start 0.30 s after the release. WFSG: fault interval located from the measured fault current as the first contiguous interval in which its 10 ms RMS exceeds $\max(0.3\,\mathrm{A}, 6\sigma)$, σ being the standard deviation of that RMS over 0.05–0.40 s, ending at the first fall below the threshold after 50 ms; reference interval from 0.05 s after the start of the record to 10 ms before the onset, split at 0.30 s; recovery windows start 0.15 s after the end of the fault interval. At 23.9–63.7 Hz a WFSG faulted interval holds between zero and eleven three-cycle windows; four trials hold none and are dropped, and 81 of 438 hold fewer than three (all 50 trials at 1671 rpm, 26 of 206 at 1800 rpm, 4 at 716 and 1 at 955 rpm).

Null and screen. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 inter-turn and inter-winding records of the WFSG. The screen flags a feature when its count of false alarms on the nine healthy trials is incompatible with the nominal rate $1 - \alpha$ at the 5 %

Table A.4: Transfer between the three alternatives with a self-referenced GBDT detector, three-cycle windows, common features

Trained on	Tested on	AUC all	AUC o.o.r.	TPR o.o.r.	FA healthy (%)
PMSG	PMSG	0.86	0.85	0.62	3
SCIG	SCIG	0.92	0.91	0.76	1
WFSG	WFSG	0.97	0.97	0.93	–
PMSG	SCIG	0.88	0.88	0.60	0
PMSG	WFSG	0.97	0.98	0.89	–
SCIG	PMSG	0.81	0.80	0.35	0
SCIG	WFSG	0.95	0.96	0.86	–
WFSG	PMSG	0.86	0.84	0.52	1
WFSG	SCIG	0.91	0.89	0.61	0
SCIG+WFSG	PMSG	0.84	0.83	0.41	0
PMSG+WFSG	SCIG	0.89	0.88	0.61	0
PMSG+SCIG	WFSG	0.97	0.97	0.88	–

Features $|z|$-referenced to the first half of each trial's own reference interval. AUC on all windows and on the out-of-reference (o.o.r.) windows; TPR: true-positive rate at 1 % false-positive rate (FPR) on the o.o.r. windows; FA: share of the healthy trials' fault-time windows above the threshold giving 1 % FPR on all negatives. The WFSG has no healthy trials, so its negatives are reference and recovery windows only and its rows are optimistic.

level (one-sided binomial test: three or more at $\alpha = 0.95$, two or more at $\alpha = 0.99$, four or more at $\alpha = 0.90$) and excludes features whose null quantile exceeds one. At $\alpha = 0.95$ it removes, at three cycles, D_2/D_1 on the SCIG; at five cycles $I_d\,1f_e$ and speed std on the PMSG and I_a h7 and D_2/D_1 on the SCIG; at eight cycles PI_d std and speed std on the PMSG and $I_q\,2f_e$, D_2/D_1 and V_{dc} std on the SCIG.

Bootstrap. 2000 resamples, seed 1. Median SDR and detectable share of a class: cluster bootstrap over the twelve tap pairs, the same resampled pairs for both machines; Wilson 95 % intervals for shares. Minimum detectable extent: operating-point trials resampled within each extent level and the limit-of-detection rule applied to each resample. The null quantile is held at its full-sample value. Decision table: 1000 resamples per suite, alternative, class and α (seed 7); a suite *meets* e_{req} when the resample probability of $\mathrm{MDE} \leq e_{\mathrm{req}}$ is at least 0.95 and *fails* when it is at most 0.05.

Transfer. Models: scikit-learn histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) and logistic regression with standardisation ($C = 1$, 3000 iterations); five-fold GroupKFold with one group per tap pair and machine, healthy trials grouped by speed. The self-referencing rule uses the windows starting before 0.62 s

(PMSG/SCIG) or 0.30 s (WFSG) as the normalising set. Metrics: area under the receiver operating characteristic curve (AUC) and true-positive rate at 1 % false-positive rate, on all windows and on the windows outside the normalising set, and the false-alarm rate of the all-negatives 1 % threshold on the fault-time windows of the healthy trials.

References

[1] M. Riera-Guasp, J. A. Antonino-Daviu, G.-A. Capolino, Advances in electrical machine, power electronic, and drive condition monitoring and fault detection: State of the art, IEEE Trans Ind Electron 62 (2015) 1746–1759. https://doi.org/10.1109/TIE.2014.2375853.

[2] A. Gandhi, T. Corrigan, L. Parsa, Recent advances in modeling and online detection of stator interturn faults in electrical motors, IEEE Trans Ind Electron 58 (2011) 1564–1575. https://doi.org/10.1109/TIE.2010.2089937.

[3] P. J. Tavner, J. Xiang, F. Spinato, Reliability analysis for wind turbines, Wind Energy 10 (2007) 1–18. https://doi.org/10.1002/we.204.

[4] J. Carroll, A. McDonald, D. McMillan, Failure rate, repair time and unscheduled O&M cost analysis of offshore wind turbines, Wind Energy 19 (2016) 1107–1119. https://doi.org/10.1002/we.1887.

[5] W. T. Thomson, M. Fenger, Current signature analysis to detect induction motor faults, IEEE Ind Appl Mag 7 (2001) 26–34. https://doi.org/10.1109/2943.930988.

[6] G. B. Kliman, W. J. Premerlani, R. A. Koegl, D. Hoeweler, A new approach to on-line turn fault detection in AC motors, in: IAS '96. Conference Record of the 1996 IEEE Industry Applications Conference, volume 1, 1996, pp. 687–693. https://doi.org/10.1109/IAS.1996.557113.

[7] S. B. Lee, R. M. Tallam, T. G. Habetler, A robust, on-line turn-fault detection technique for induction machines based on monitoring the sequence component impedance matrix, IEEE Trans Power Electron 18 (2003) 865–872. https://doi.org/10.1109/TPEL.2003.810848.

[8] S. Nandi, H. A. Toliyat, X. Li, Condition monitoring and fault diagnosis of electrical motors – a review, IEEE Trans Energy Convers 20 (2005) 719–729. https://doi.org/10.1109/TEC.2005.847955.

[9] A. Bellini, F. Filippetti, C. Tassoni, G.-A. Capolino, Advances in diagnostic techniques for induction machines, IEEE Trans Ind Electron 55 (2008) 4109–4126. https://doi.org/10.1109/TIE.2008.2007527.

[10] R. M. Tallam, T. G. Habetler, R. G. Harley, Stator winding turn-fault detection for closed-loop induction motor drives, IEEE Trans Ind Appl 39 (2003) 720–724. https://doi.org/10.1109/TIA.2003.811784.

[11] S. M. A. Cruz, A. J. M. Cardoso, Diagnosis of stator inter-turn short circuits in DTC induction motor drives, IEEE Trans Ind Appl 40 (2004) 1349–1360. https://doi.org/10.1109/TIA.2004.834012.

[12] H. Polinder, F. F. A. van der Pijl, G.-J. de Vilder, P. J. Tavner, Comparison of direct-drive and geared generator concepts for wind turbines, IEEE Trans Energy Convers 21 (2006) 725–733. https://doi.org/10.1109/TEC.2006.875476.

[13] H. Polinder, J. A. Ferreira, B. B. Jensen, A. B. Abrahamsen, K. Atallah, R. A. McMahon, Trends in wind turbine generator systems, IEEE J Emerg Sel Top Power Electron 1 (2013) 174–185. https://doi.org/10.1109/JESTPE.2013.2280428.

[14] W. Yang, P. J. Tavner, C. J. Crabtree, Y. Feng, Y. Qiu, Wind turbine condition monitoring: technical and commercial challenges, Wind Energy 17 (2014) 673–693. https://doi.org/10.1002/we.1508.

[15] M. Basseville, On fault detectability and isolability, Eur J Control 7 (2001) 625–637. https://doi.org/10.3166/ejc.7.625-637.

[16] M. Blanke, M. Kinnaert, J. Lunze, M. Staroswiecki, Diagnosis and Fault-Tolerant Control, 3 ed., Springer, Berlin, Heidelberg, 2016. https://doi.org/10.1007/978-3-662-47943-8.

[17] M. Krysander, E. Frisk, Sensor placement for fault diagnosis, IEEE Trans Syst Man Cybern A Syst Hum 38 (2008) 1398–1410. https://doi.org/10.1109/TSMCA.2008.2003968.

[18] E. Frisk, M. Krysander, J. Åslund, Sensor placement for fault isolation in linear differential-algebraic systems, Automatica 45 (2009) 364–371. https://doi.org/10.1016/j.automatica.2008.08.013.

[19] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, A. K. Nandi, Applications of machine learning to machine fault diagnosis: A review and roadmap, Mech Syst Signal Process 138 (2020) 106587. https://doi.org/10.1016/j.ymssp.2019.106587.

[20] S. J. Pan, Q. Yang, A survey on transfer learning, IEEE Trans Knowl Data Eng 22 (2010) 1345–1359. https://doi.org/10.1109/TKDE.2009.191.

[21] J. Coble, J. W. Hines, Identifying suitable degradation parameters for individual-based prognostics, in: Diagnostics and Prognostics of Engineering Systems: Methods and Techniques, IGI Global, Hershey, PA, 2013, pp. 135–150. https://doi.org/10.4018/978-1-4666-2095-7.ch007.

[22] A. Saxena, J. Celaya, E. Balaban, K. Goebel, B. Saha, S. Saha, M. Schwabacher, Metrics for evaluating performance of prognostic techniques, in: 2008 International Conference on Prognostics and Health Management, 2008, pp. 1–17. https://doi.org/10.1109/PHM.2008.4711436.

[23] M. Basseville, I. V. Nikiforov, Detection of Abrupt Changes: Theory and Application, Prentice Hall, Englewood Cliffs, NJ, 1993.

[24] R. N. Tominaga, S. S. Barbosa, L. A. Sousa, A. d. S. Lunardi, R. V. Rocha, S. L. Ávila, B. S. Carmo, R. M. Monaro, M. B. d. C. Salles, A benchmark dataset of electrical signals from a permanent magnet synchronous generator for condition monitoring, Data Brief 62 (2025) 112040. https://doi.org/10.1016/j.dib.2025.112040.

[25] R. N. Tominaga, S. S. Barbosa, L. A. Sousa, A. d. S. Lunardi, R. V. Rocha, S. L. Ávila, B. S. Carmo, R. M. Monaro, M. B. d. C. Salles, A condition monitoring dataset based on electrical signals for a squirrel cage induction generator, Data Brief 63 (2025) 112286. https://doi.org/10.1016/j.dib.2025.112286.

[26] R. N. Tominaga, L. A. Sousa, R. V. Rocha, R. M. Monaro, S. L. Ávila, M. B. d. C. Salles, B. S. Carmo, Electrical signals dataset from fixed-speed and variable-speed synchronous generators under healthy and faulty conditions, Data Brief 57 (2024) 111018. https://doi.org/10.1016/j.dib.2024.111018.

[27] J. Zhang, G. Rizzoni, Structural analysis for diagnosability and reconfigurability, with application to electric vehicle drive system, IFAC-PapersOnLine 48 (2015) 1471–1478. https://doi.org/10.1016/j.ifacol.2015.09.732.

[28] S. Deb, K. R. Pattipati, V. Raghavan, M. Shakeri, R. Shrestha, Multi-signal flow graphs: a novel approach for system testability analysis and fault diagnosis, in: Proceedings of AUTOTESTCON '94, 1994, pp. 361–373. https://doi.org/10.1109/AUTEST.1994.381596.

[29] W. R. Simpson, J. W. Sheppard, System Test and Diagnosis, Springer, Boston, MA, 1994. https://doi.org/10.1007/978-1-4615-2702-2.

[30] R. Isermann, Model-based fault-detection and diagnosis – status and applications, Annual Reviews in Control 29 (2005) 71–85. https://doi.org/10.1016/j.arcontrol.2004.12.002.

[31] V. Venkatasubramanian, R. Rengaswamy, K. Yin, S. N. Kavuri, A review of process fault detection and diagnosis: Part I: Quantitative model-based methods, Comput Chem Eng 27 (2003) 293–311. https://doi.org/10.1016/S0098-1354(02)00160-6.

[32] S. Cheng, M. H. Azarian, M. G. Pecht, Sensor systems for prognostics and health management, Sensors 10 (2010) 5774–5797. https://doi.org/10.3390/s100605774.

[33] R. M. Tallam, T. G. Habetler, R. G. Harley, Transient model for induction machines with stator winding turn faults, IEEE Trans Ind Appl 38 (2002) 632–637. https://doi.org/10.1109/TIA.2002.1003411.

[34] P. Neti, S. Nandi, Stator interturn fault detection of synchronous machines using field current and rotor search-coil voltage signature analysis, IEEE Trans Ind Appl 45 (2009) 911–920. https://doi.org/10.1109/TIA.2009.2018905.

[35] J.-C. Urresty, J.-R. Riba, L. Romeral, Application of the zero-sequence voltage component to detect stator winding inter-turn faults in PMSMs, Electr Power Syst Res 89 (2012) 38–44. https://doi.org/10.1016/j.epsr.2012.02.012.

[36] H. T. Canseven, M. Ercire, M. Cömert, A. Ünsal, N. Sarma, Advanced fault detection of permanent magnet faults in offshore wind turbine generators using finite element analysis and deep transfer learning, Machines 14 (2026b) 665. https://doi.org/10.3390/machines14060665.

[37] E. Ş. Sadık, H. T. Canseven, Onset-aware alarm-triggered fault diagnosis in collaborative robots, IEEE Access 14 (2026) 115596–115608. https://doi.org/10.1109/ACCESS.2026.3717458.

[38] W. Jung, S.-H. Yun, Y.-S. Lim, S. Cheong, Y.-H. Park, Vibration and current dataset of three-phase permanent magnet synchronous motors with stator faults, Data Brief 47 (2023) 108952. https://doi.org/10.1016/j.dib.2023.108952.

[39] B. Vaseghi, N. Takorabet, F. Meibody-Tabar, Fault analysis and parameter identification of permanent-magnet motors by the finite-element method, IEEE Trans Magn 45 (2009) 3290–3295. https://doi.org/10.1109/TMAG.2009.2022156.

[40] F. Tao, H. Zhang, A. Liu, A. Y. C. Nee, Digital twin in industry: State-of-the-art, IEEE Trans Ind Inform 15 (2019) 2405–2415. https://doi.org/10.1109/TII.2018.2873186.

[41] I. Y. Tumer, R. B. Stone, Mapping function to failure mode during component development, Res Eng Des 14 (2003) 25–33. https://doi.org/10.1007/s00163-002-0024-y.

[42] R. B. Stone, I. Y. Tumer, M. Van Wie, The function-failure design method, J Mech Des 127 (2005) 397–407. https://doi.org/10.1115/1.1862678.

[43] T. Kurtoglu, I. Y. Tumer, D. C. Jensen, A functional failure reasoning methodology for evaluation of conceptual system architectures, Res Eng Des 21 (2010) 209–234. https://doi.org/10.1007/s00163-010-0086-1.

[44] T. W. Simpson, Product platform design and customization: Status and promise, Artif Intell Eng Des Anal Manuf 18 (2004) 3–20. https://doi.org/10.1017/S0890060404040028.

[45] R. V. Rocha, L. A. Sousa, R. M. Monaro, Experimental platform for evaluation of full-scale variable-speed wind generators under stator winding faults, IEEE Trans Energy Convers 38 (2023) 800–812. https://doi.org/10.1109/TEC.2022.3230260.

[46] E. Gaertner, J. Rinker, L. Sethuraman, F. Zahle, B. Anderson, G. Barter, N. Abbas, F. Meng, P. Bortolotti, W. Skrzypinski, G. Scott, R. Feil, H. Bredmose, K. Dykes, M. Shields, C. Allen, A. Viselli, IEA Wind TCP Task 37: Definition of the IEA 15-Megawatt Offshore Reference Wind Turbine, Technical Report NREL/TP-5000-75698, National Renewable Energy Laboratory, 2020. https://doi.org/10.2172/1603478.

[47] J. H. Friedman, Greedy function approximation: a gradient boosting machine, Ann Stat 29 (2001) 1189–1232. https://doi.org/10.1214/aos/1013203451.